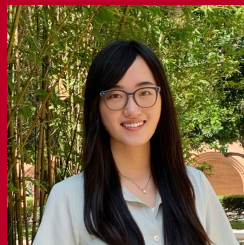


Carnegie Mellon University

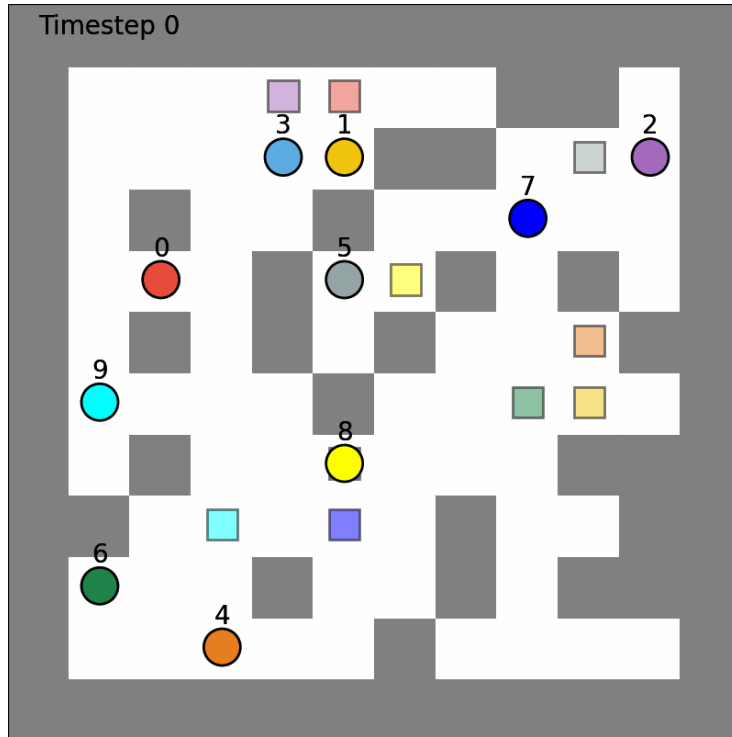
Solving Multi-Agent Target Assignment and Path Finding with a Single Constraint Tree

Yimin Tang, Zhongqiang Ren, Jiaoyang Li, Katia Sycara



Multi Agent Path Finding Problem (MAPF)

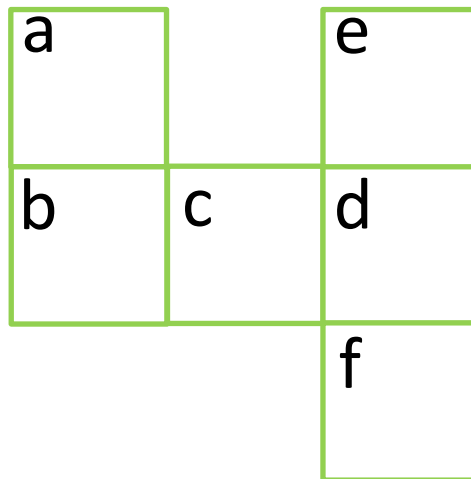
Multi Agent Path Finding Problem (MAPF)



<https://www.machinedesign.com/mechanical-motion-systems/article/21835788/changing-the-future-of-warehouses-with-amazon-robots>
<https://apps.boschrexroth.com/microsites/ctrlx-automation/en/portfolio/ctrlx-flow/>
<https://jiaoyangli.me>

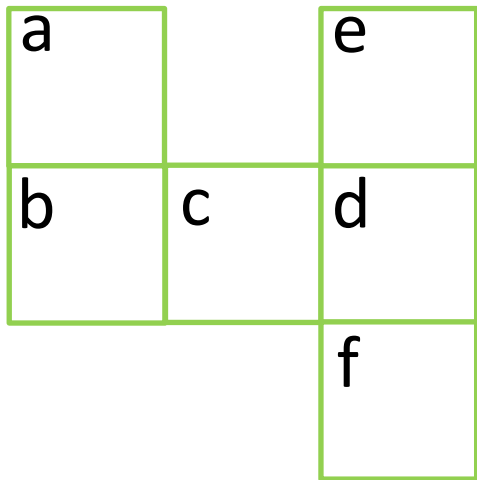
Multi Agent Path Finding Problem (MAPF)

- Input:
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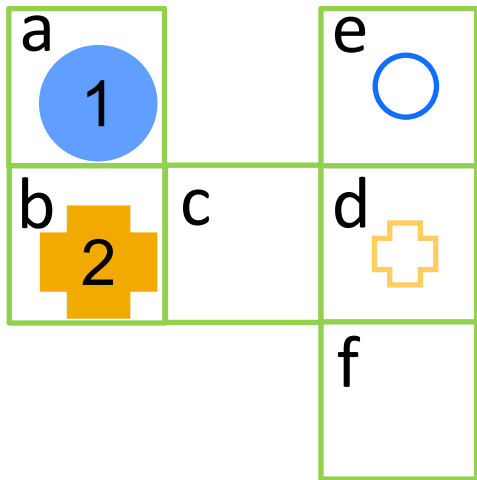
Multi Agent Path Finding Problem (MAPF)

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 - multiple agents (each with one start and one goal location)



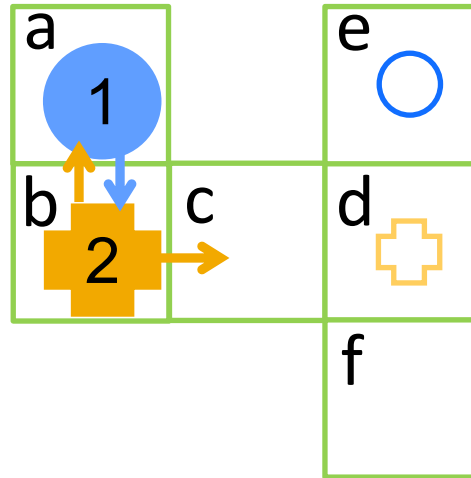
Multi Agent Path Finding Problem (MAPF)

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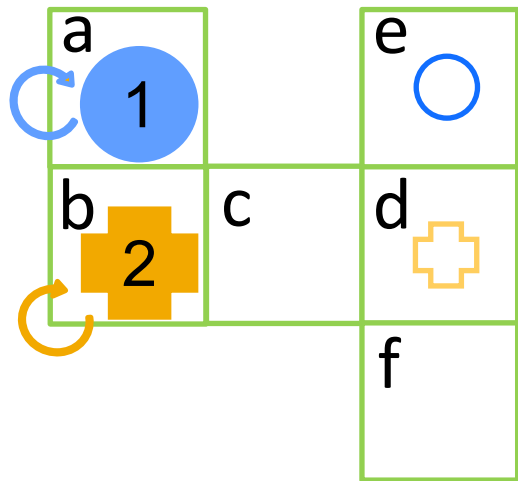
Multi Agent Path Finding Problem (MAPF)

- Action:
 - Move: move to a neighboring location



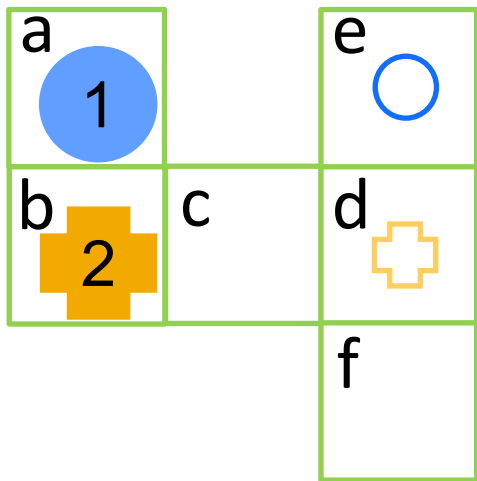
Multi Agent Path Finding Problem (MAPF)

- Action:
 - Move: move to a neighboring location
 - Wait: wait at its current location



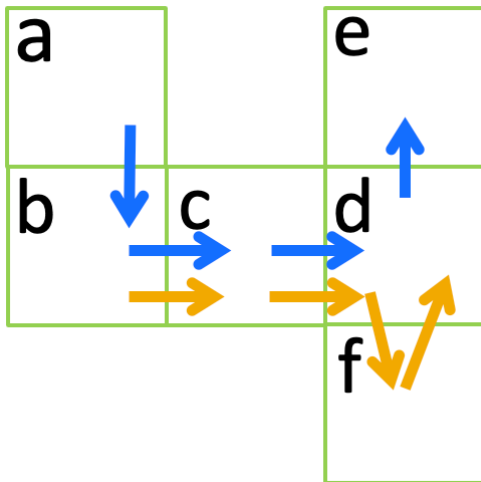
Multi Agent Path Finding Problem (MAPF)

- Output:
 - Collision-free paths for all agents that minimize flowtime
 - Flowtime: sum of the arrival time for all agents



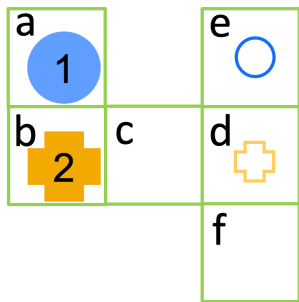
Multi Agent Path Finding Problem (MAPF)

- Output:
 - MAPF is NP-hard problem
 - Flowtime: 4 (Blue $\rightarrow$ e) + 4 (Yellow $\rightarrow$ d) = 8



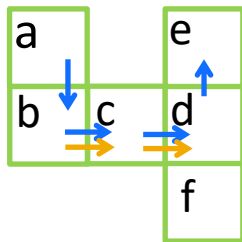
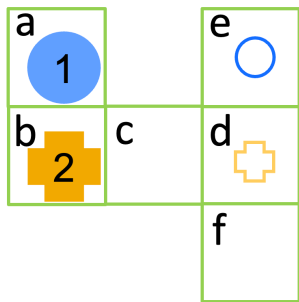
Conflict-Based Search (CBS)

Optimal method for MAPF



Conflict-Based Search (CBS)

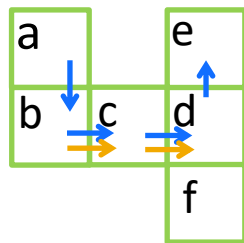
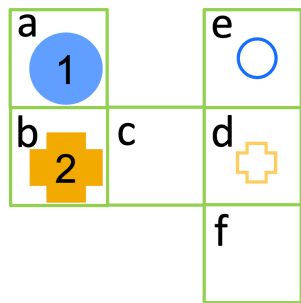
Optimal method for MAPF



Conflict at location **d**
and timestep **3**

Conflict-Based Search (CBS)

Optimal method for MAPF



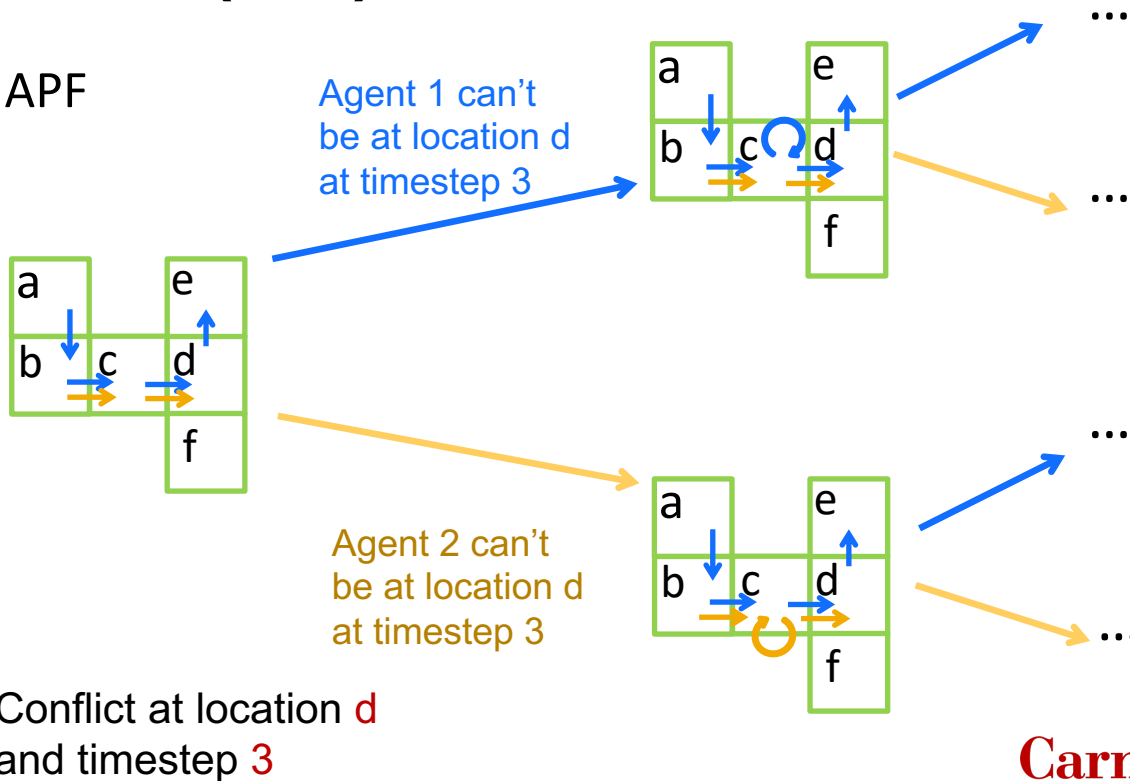
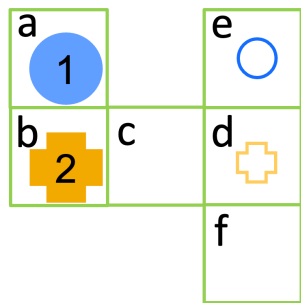
Agent 1 can't
be at location d
at timestep 3

Agent 2 can't
be at location d
at timestep 3

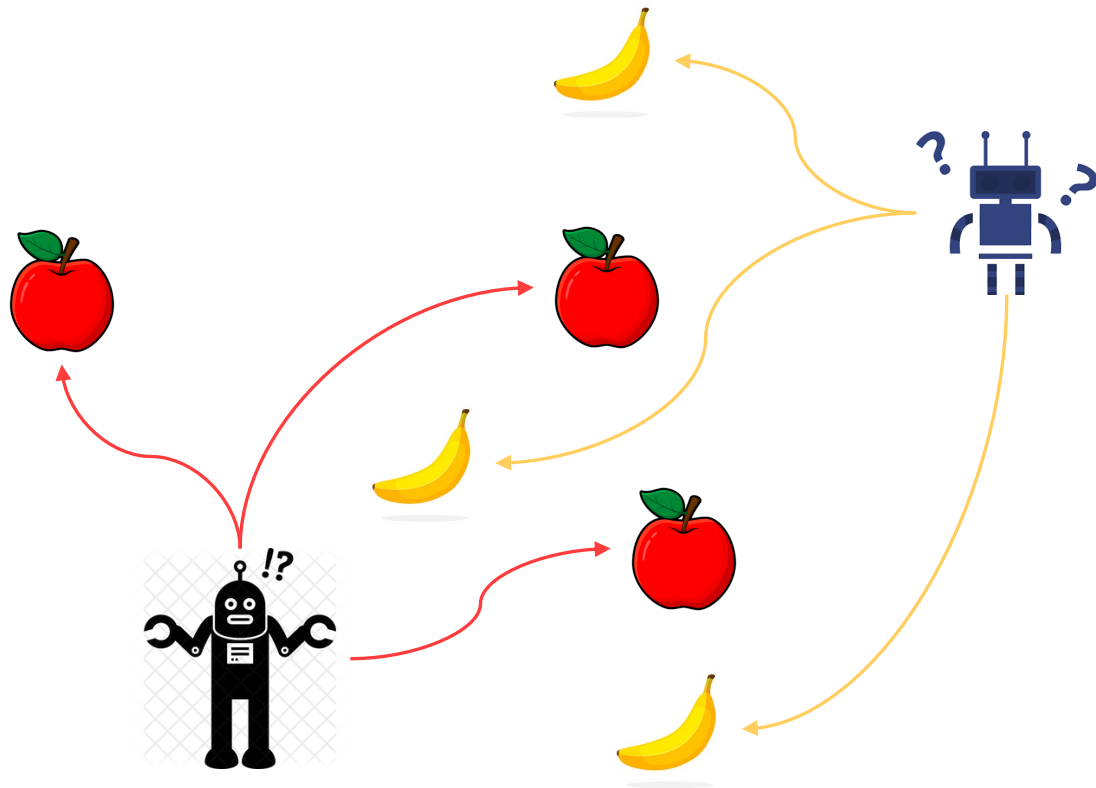
Conflict at location **d**
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Conflict-Based Search (CBS)

Optimal method for MAPF

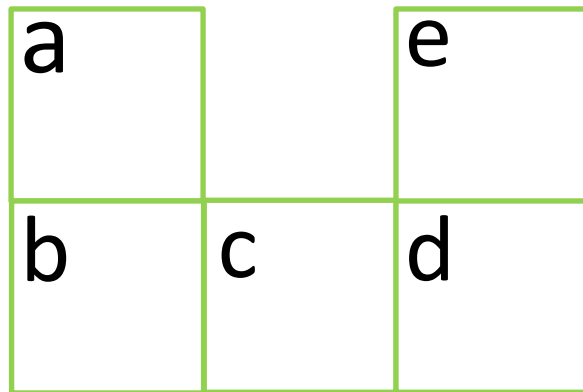


Target-Assignment and Path-Finding Problem (TAPF)



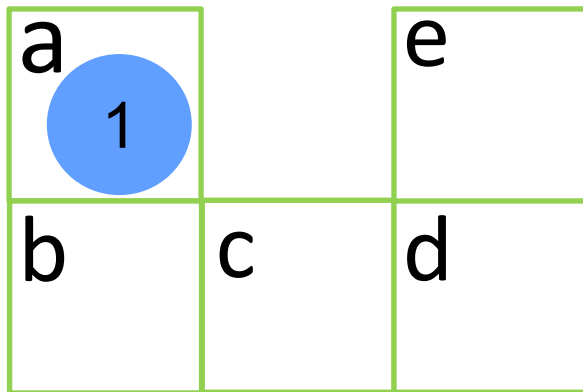
Target-Assignment and Path-Finding Problem (TAPF)

- Input:
 - a known graph



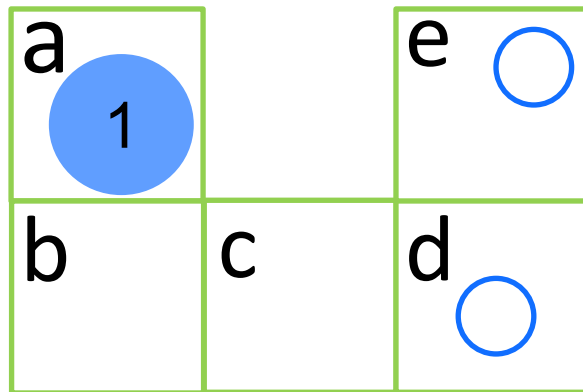
Target-Assignment and Path-Finding Problem (TAPF)

- Input:
 - a known graph
 - agents (each with a start location and a goal location set)



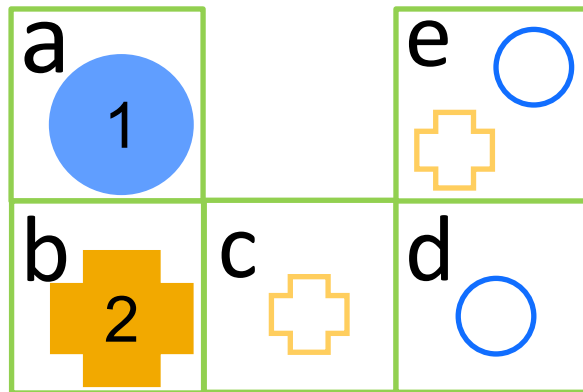
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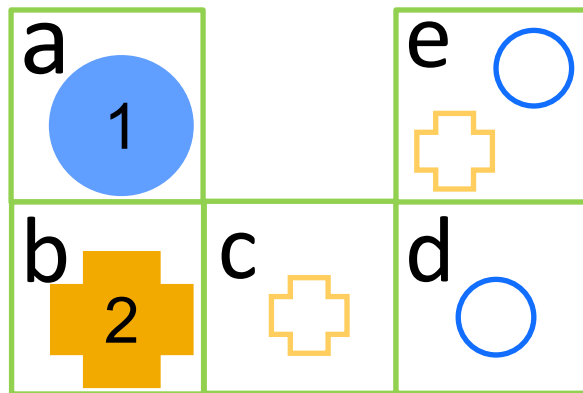
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Target-Assignment and Path-Finding Problem (TAPF)

- Output:
 - Agent only needs go to **one** of its target location set
 - **Collision-free paths** for all agents and **minimize flowtime**



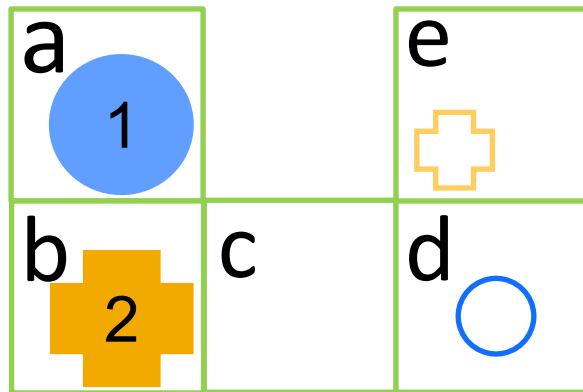
Target-Assignment and Path-Finding Problem (TAPF)

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Task Assignment (TA):

1 → d

2 → e



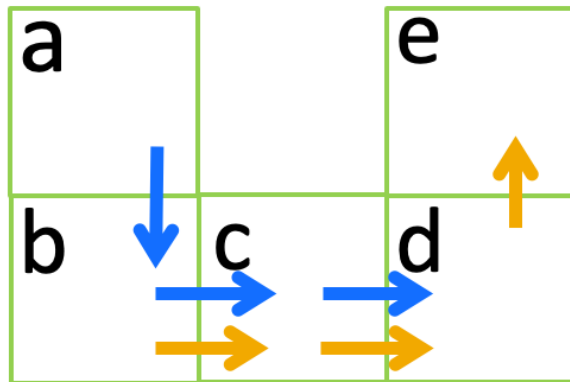
Target-Assignment and Path-Finding Problem (TAPF)

- Output:
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 - **Collision-free paths** for all agents

Task Assignment (TA):

1 → d

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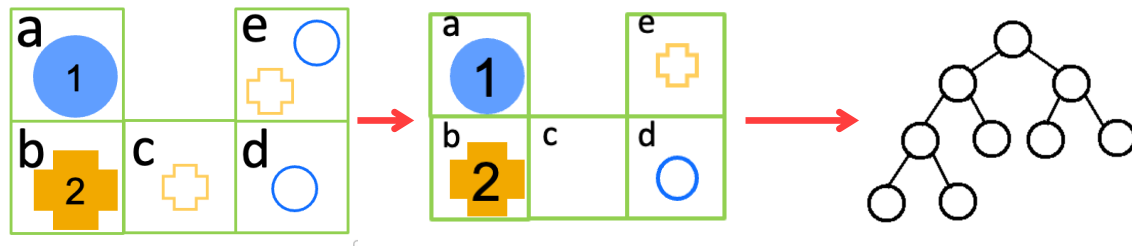


Conflict-Based Search with Target Assignment (CBS-TA)

Conflict-Based Search with Target Assignment (CBS-TA)

Optimal method for TAPF

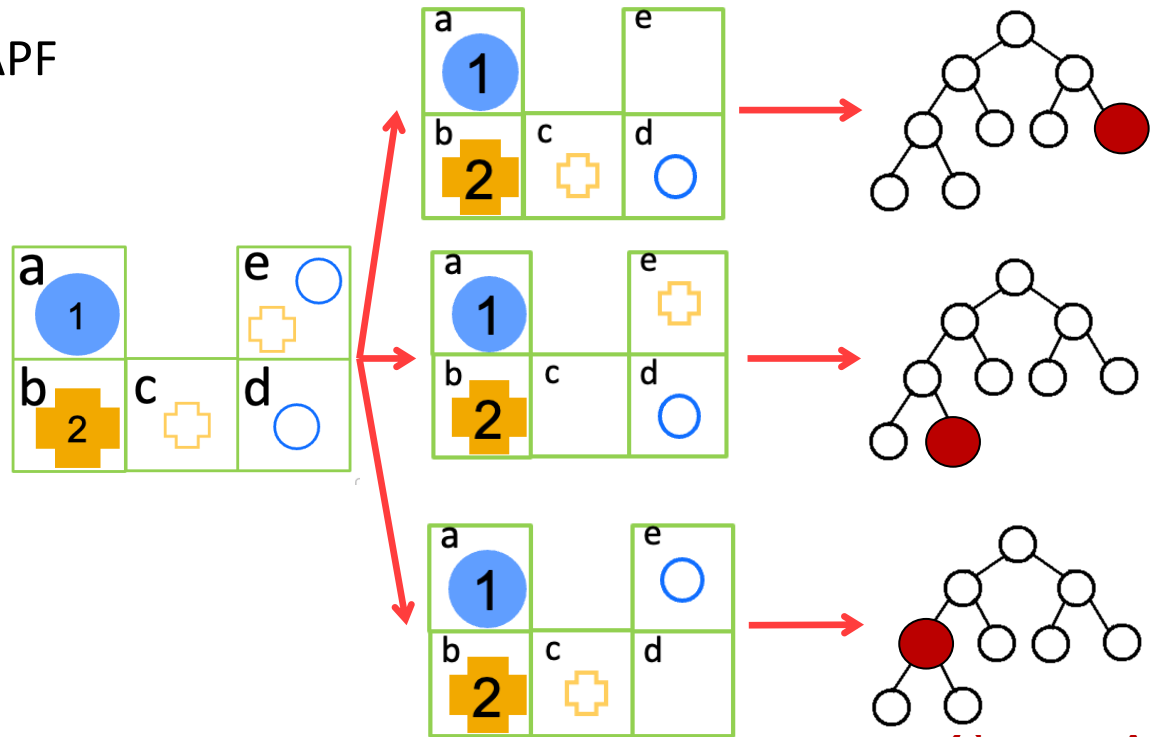
- When Target Assignment (TA) is given, TAPF becomes MAPF



Conflict-Based Search with Target Assignment (CBS-TA)

Optimal method for TAPF

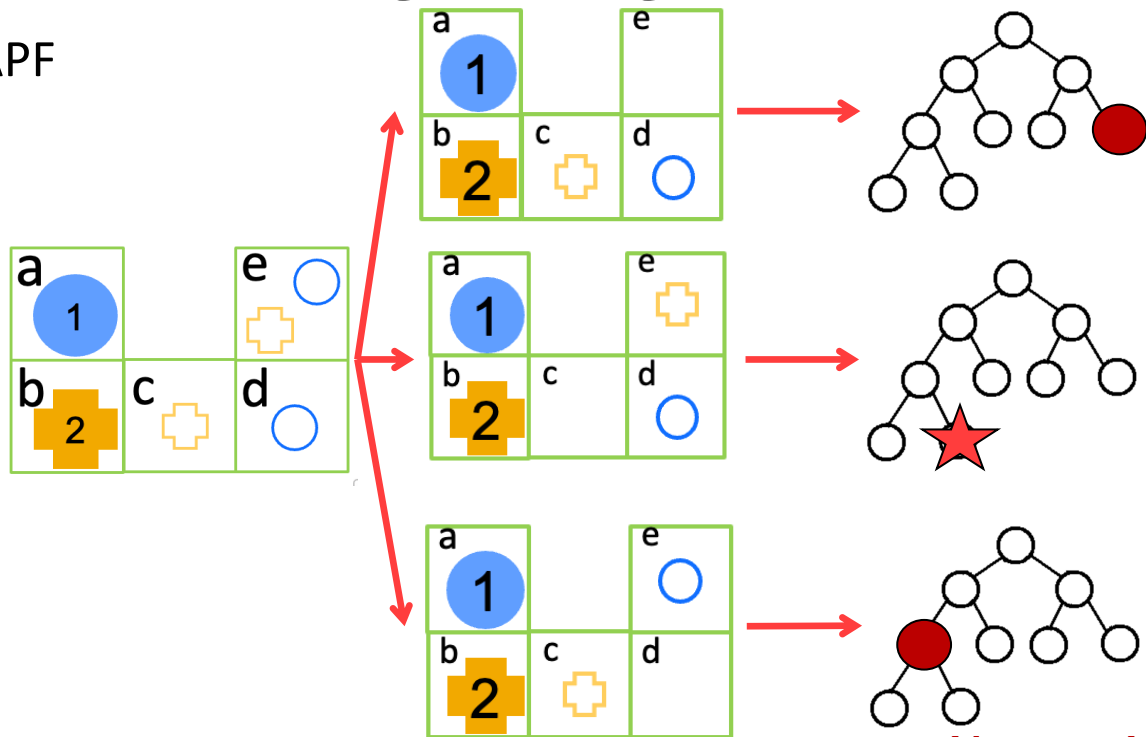
- When Target Assignment (TA) is given, TAPF becomes MAPF
- Each different Constraint Tree (CBS Tree) → one optimal answer by CBS



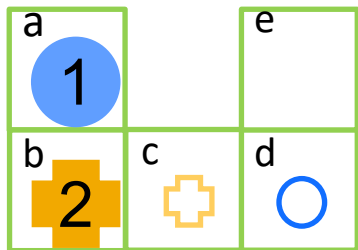
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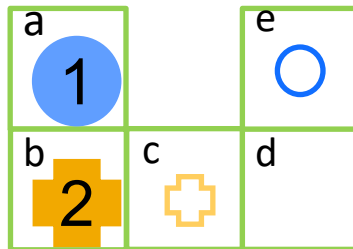
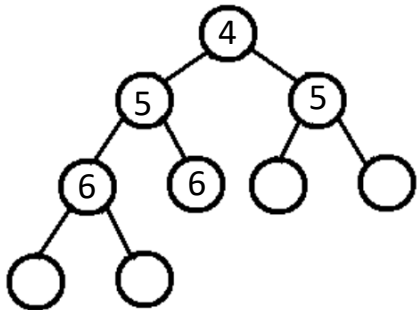
- When Target Assignment (TA) is given, TAPF becomes MAPF
- Each different Constraint Tree (CBS Tree) → one optimal answer by CBS
- From the **forest**, get the best one



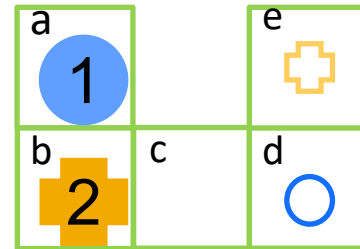
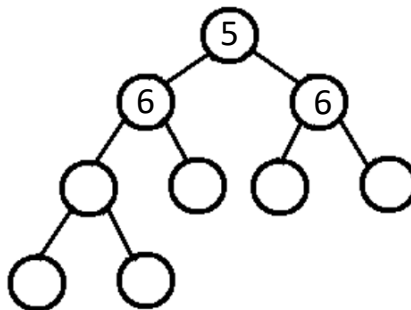
Conflict-Based Search with Target Assignment (CBS-TA)



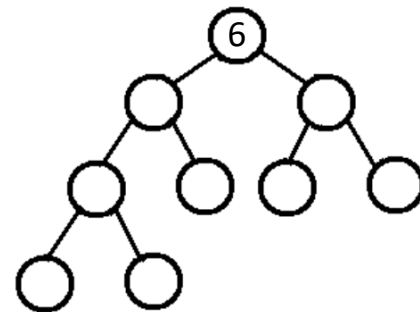
Cost:4



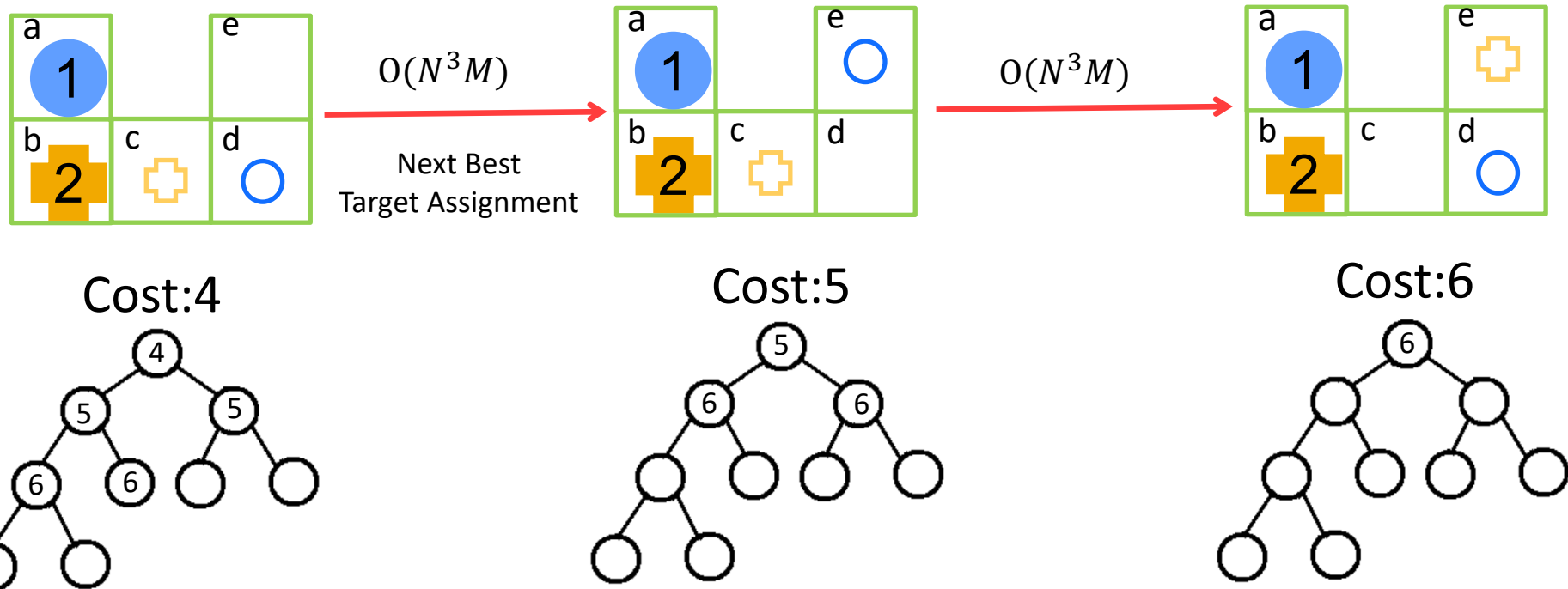
Cost:5



Cost:6



Conflict-Based Search with Target Assignment (CBS-TA)



CBS-TA issues

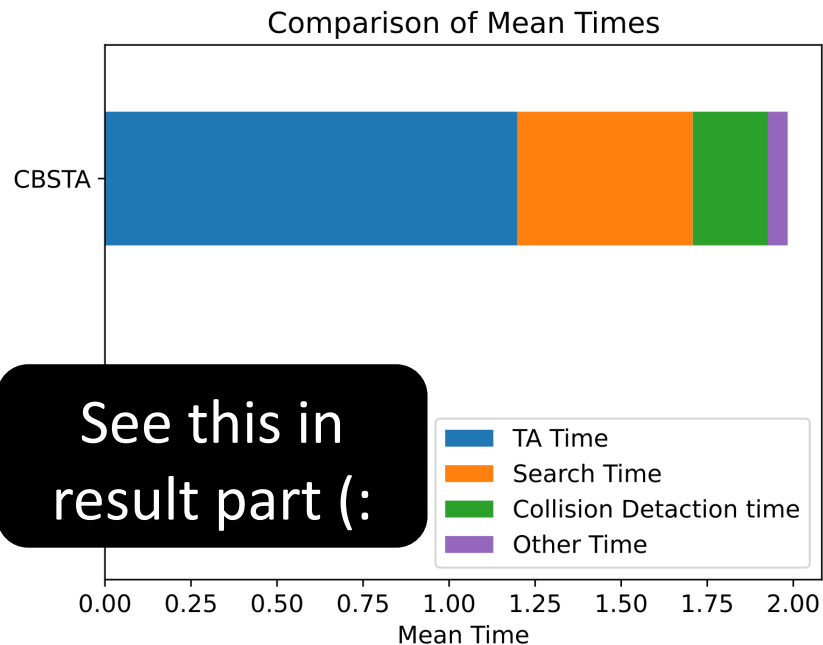
- Target Assignment (TA) is time consuming

CBS-TA issues

- Target Assignment (TA) is time consuming
- $O(N^3M)$
- N is #agent (max: 60)
- M is #all targets (max: $60 \cdot 80 = 4800$)
- $N^3M = 10^9$

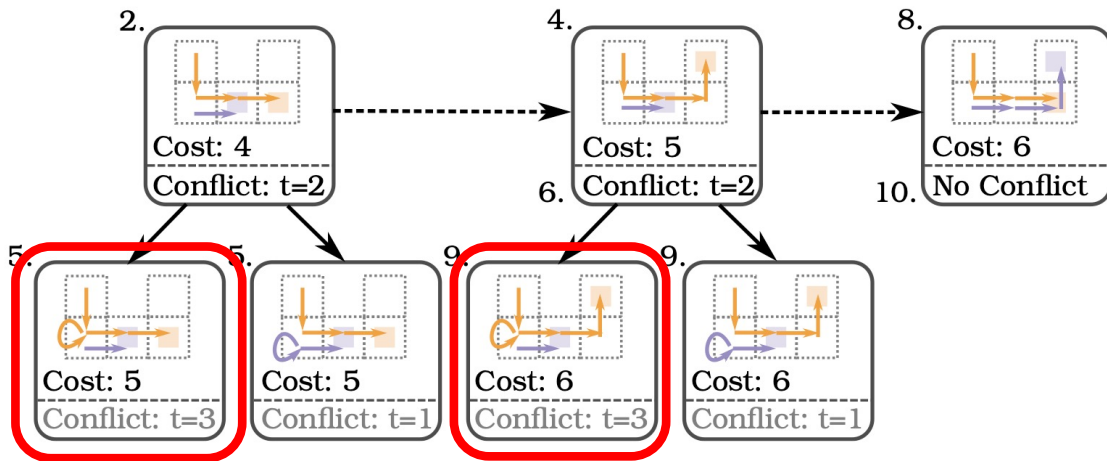
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CBS-TA issues

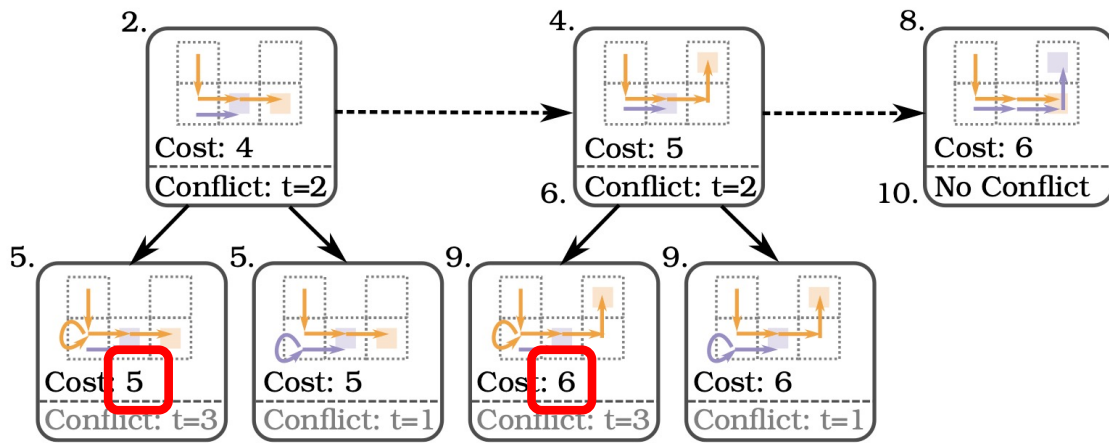
- Generate multiple nodes with the **same constraint set**



(d) CBS-TA search forest

CBS-TA issues

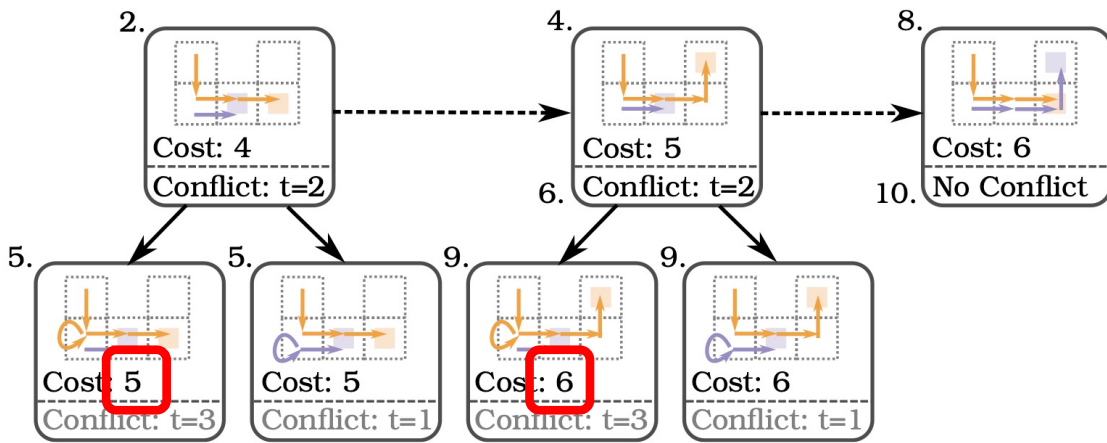
- Generate multiple nodes with the **same constraint set**
- But under the same constraint set, their **satisfied optimal solution cost are different**



(d) CBS-TA search forest

CBS-TA issues

- Generate multiple nodes with the **same constraint set**
- But under the same constraint set, their **satisfied optimal solution cost are different**
- Only one of them could become TAPF optimal solution
- Exist some optimization space



(d) CBS-TA search forest

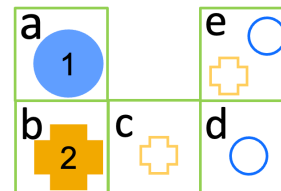
Incremental Target Assignment Conflict-Based Search (ITA-CBS)

Incremental Target Assignment Conflict-Based Search (ITA-CBS)

- ITA-CBS is optimal

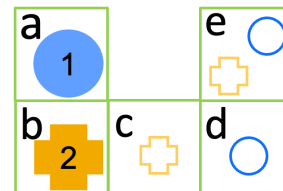
Incremental Target Assignment Conflict-Based Search (ITA-CBS)

- ITA-CBS is **optimal**
- ITA-CBS Tree Node:



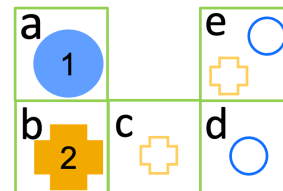
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Incremental Target Assignment Conflict-Based Search (ITA-CBS)

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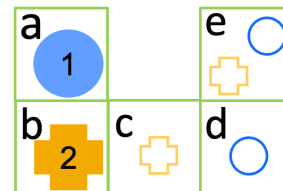
- ITA-CBS Tree Node:

- Constraint Set Ω

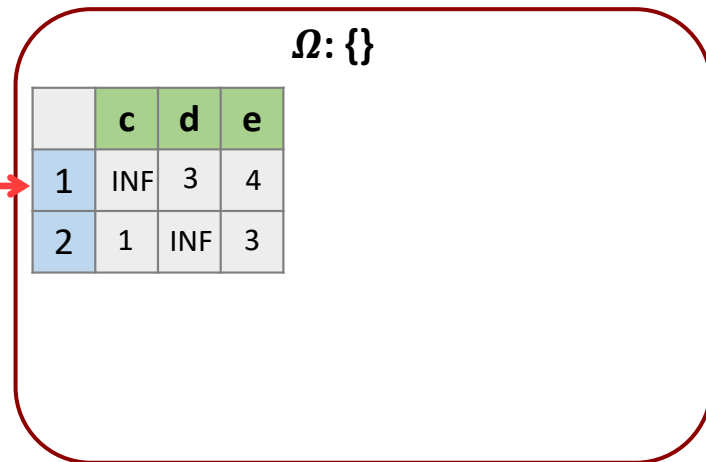
$\Omega: \{\}$

Incremental Target Assignment Conflict-Based Search (ITA-CBS)

- ITA-CBS is **optimal**

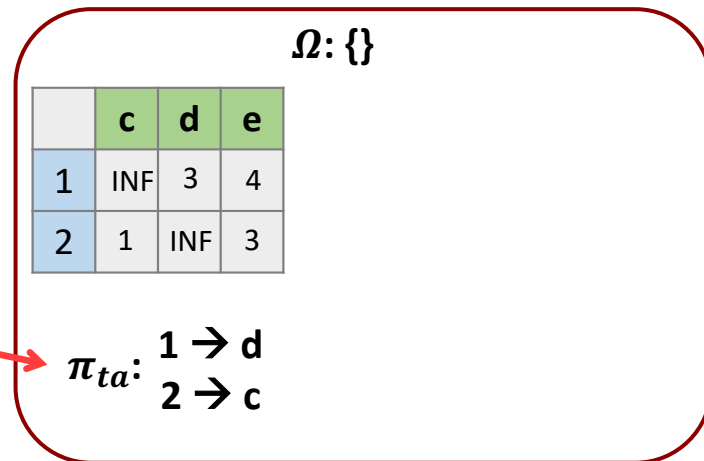
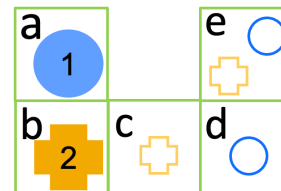


- ITA-CBS Tree Node:
 - Constraint Set Ω
 - Shortest Path Cost Matrix M under Ω



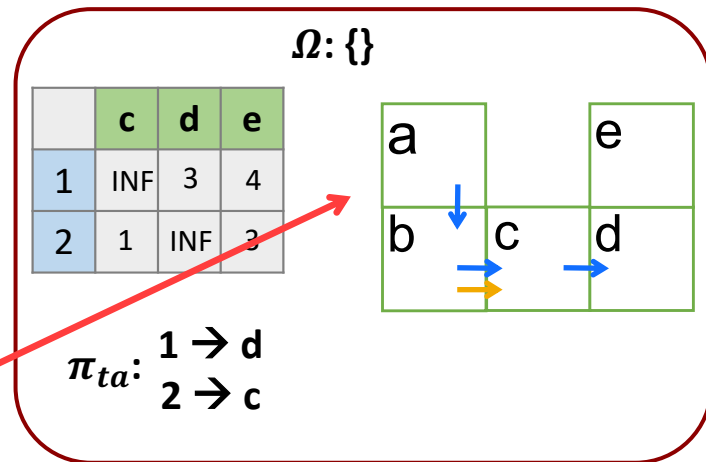
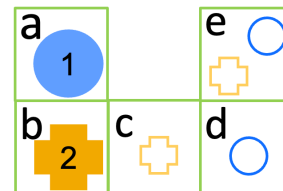
Incremental Target Assignment Conflict-Based Search (ITA-CBS)

- ITA-CBS is **optimal**
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 - Shortest Path Cost Matrix M under Ω
 - Optimal Target Assignment from M



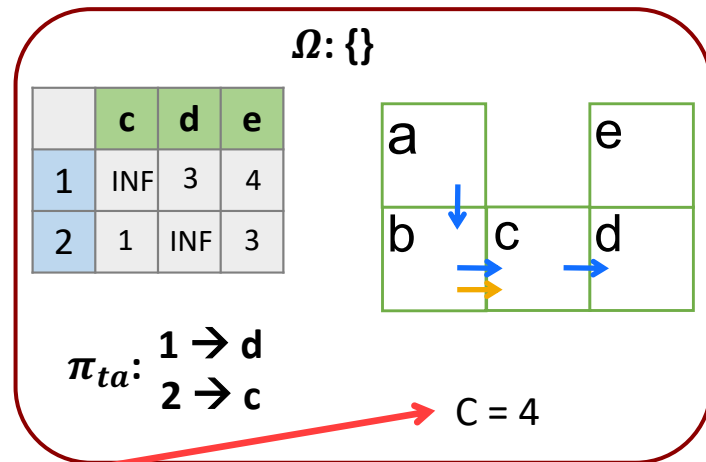
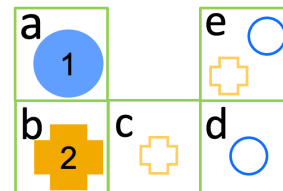
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Incremental Target Assignment Conflict-Based Search (ITA-CBS)

- ITA-CBS is **optimal**
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 - Flowtime C

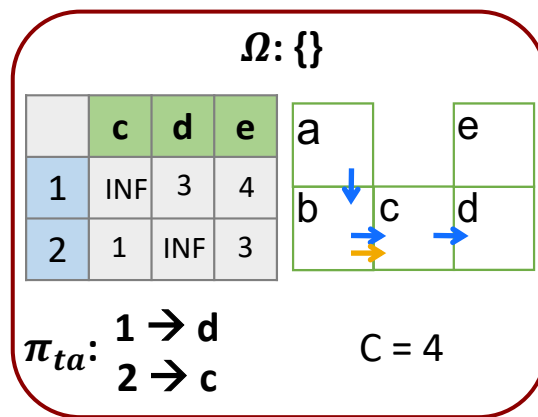


ITA-CBS (Ours)

It only has a **single tree** rather than a **forest** in CBS-TA

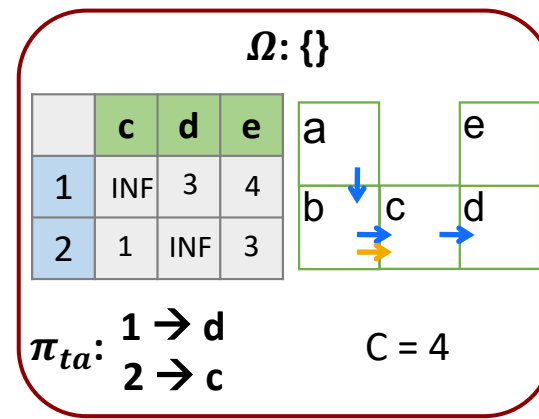
ITA-CBS (Ours)

1. Create a constraint tree root node and verify



ITA-CBS (Ours)

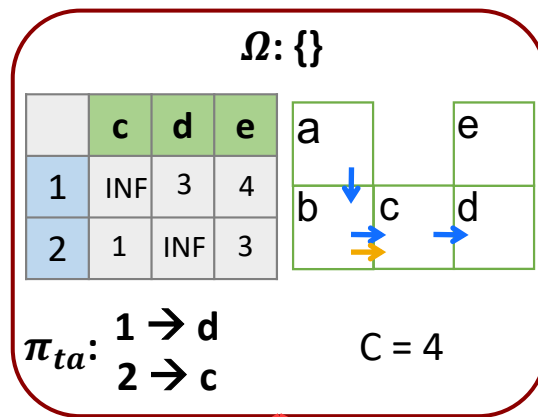
1. Create a constraint tree root node and verify
2. If not, use the first conflict to generate two child nodes



Conflict at (c, time: 2)

ITA-CBS (Ours)

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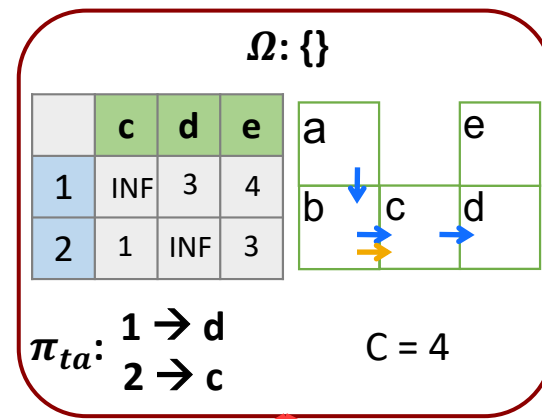
Conflict at (c, time: 2)

$\Omega: \{(\text{Blue}, c, 2)\}$

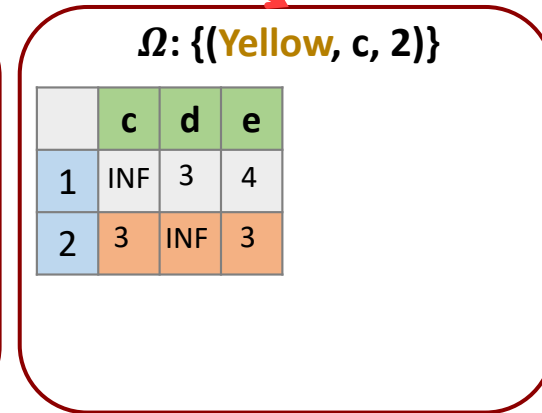
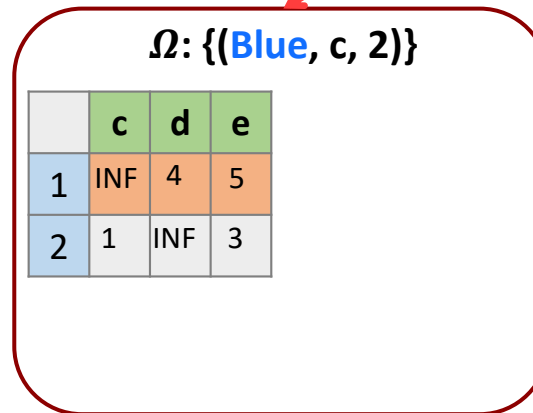
$\Omega: \{(\text{Yellow}, c, 2)\}$

ITA-CBS (Ours)

1. Create a constraint tree root node and verify
2. If not, use the first conflict to generate two child nodes
3. Find shortest path, update cost matrix and TA

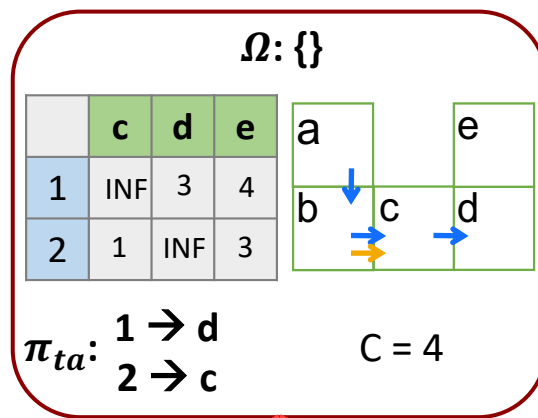


Conflict at (c, time: 2)

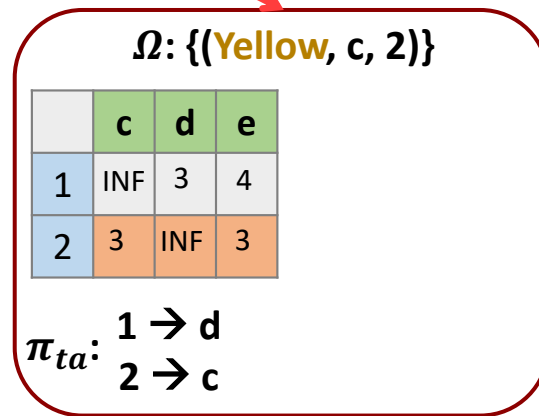
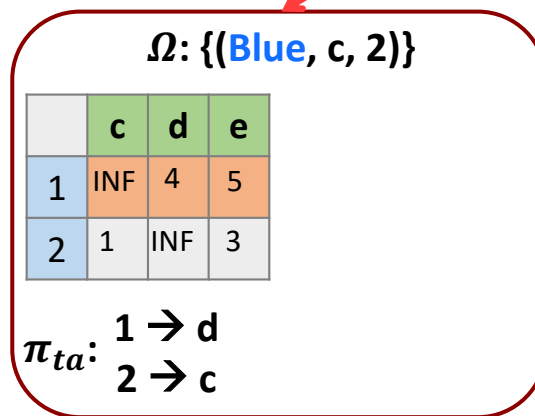


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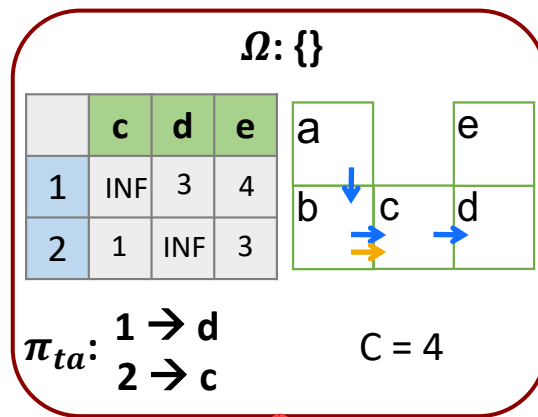


Conflict at (c, time: 2)

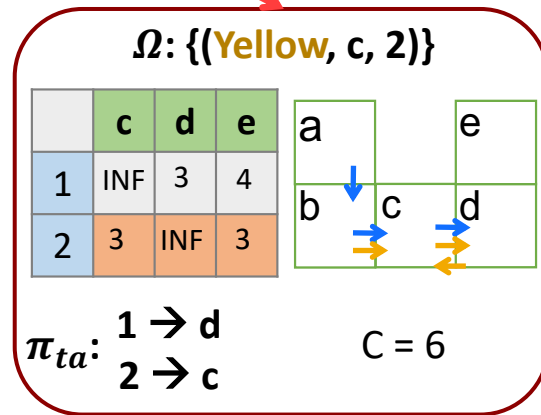
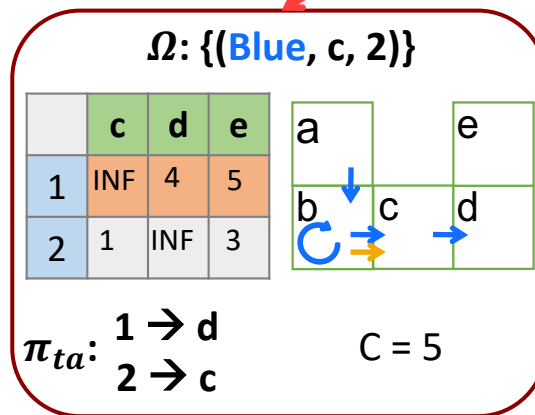


ITA-CBS (Ours)

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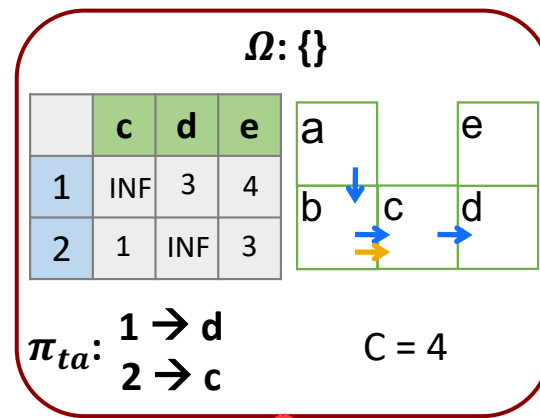


Conflict at (c, time: 2)

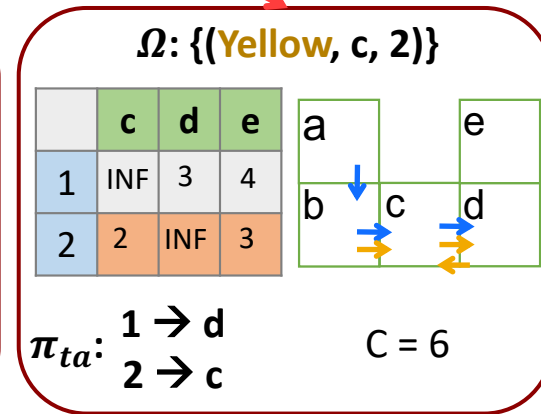
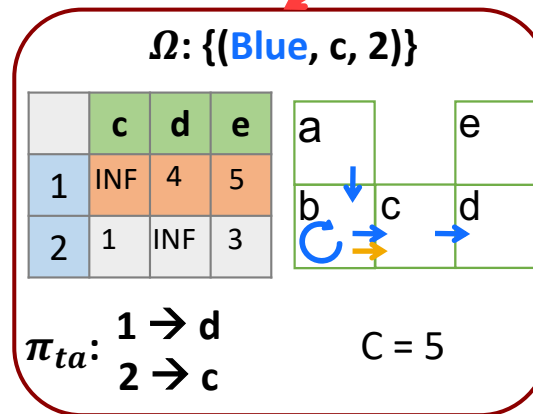


ITA-CBS (Ours)

1. Create a constraint tree root node and verify
2. If not, use the first conflict to generate two child nodes
3. Find shortest path, update cost matrix and TA
4. Get solution based on TA
5. Sort all new nodes and select smallest node, goto 2



Conflict at (c, time: 2)



ITA-CBS (Ours)

- Each node will need to re-run Target Assignment (TA) algorithm

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ITA-CBS (Ours)

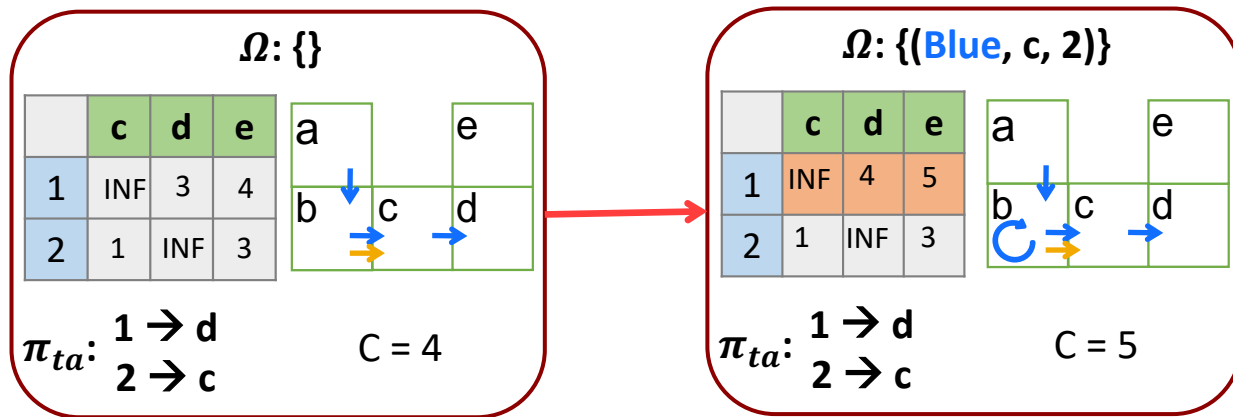
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- Optimization:

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 - Child node only needs to update one row in cost matrix

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 - Dynamic Hungarian Algorithm

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- Target Assignment algorithm : Hungarian algorithm needs $O(N^2M)$
- Optimization:
 - Child node only needs to update one row in cost matrix
 - Dynamic Hungarian Algorithm
 - *only needs $O(NM)$ in ITA-CBS*

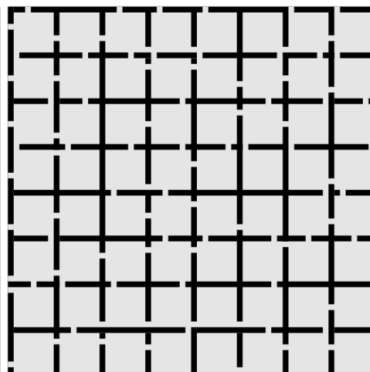
Experiment Results

Tang, Yimin, et al. "Solving Multi-Agent Target Assignment and Path Finding with a Single Constraint Tree." in submission

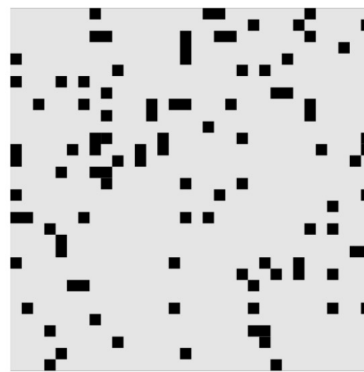
Experiment Results



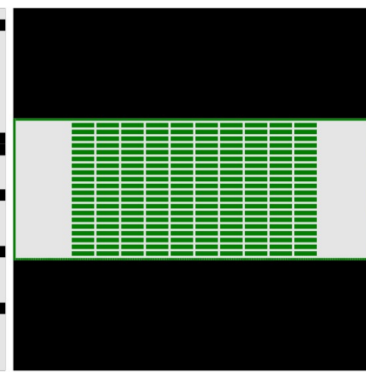
(a) den312d



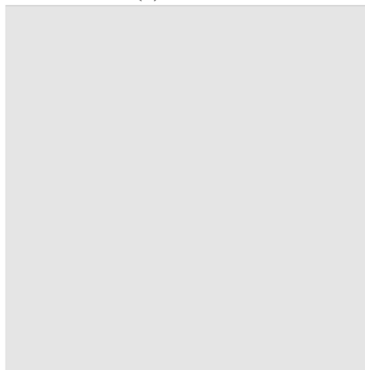
(b) room-64-64-8



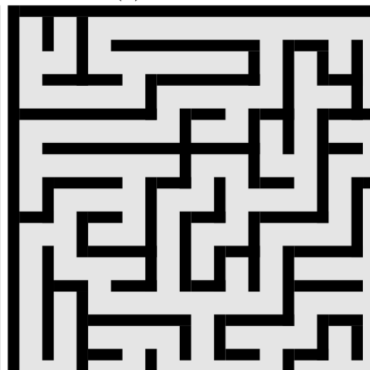
(a) random-32-32-10



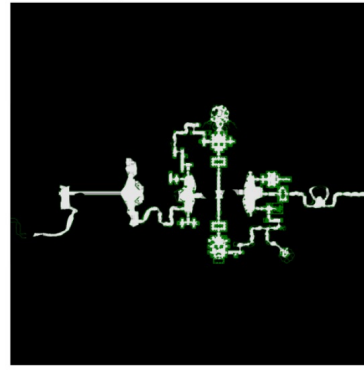
(b) warehouse-10-20-10-2-1



(c) empty-32-32



(d) maze-32-32-2



(c) orz900d



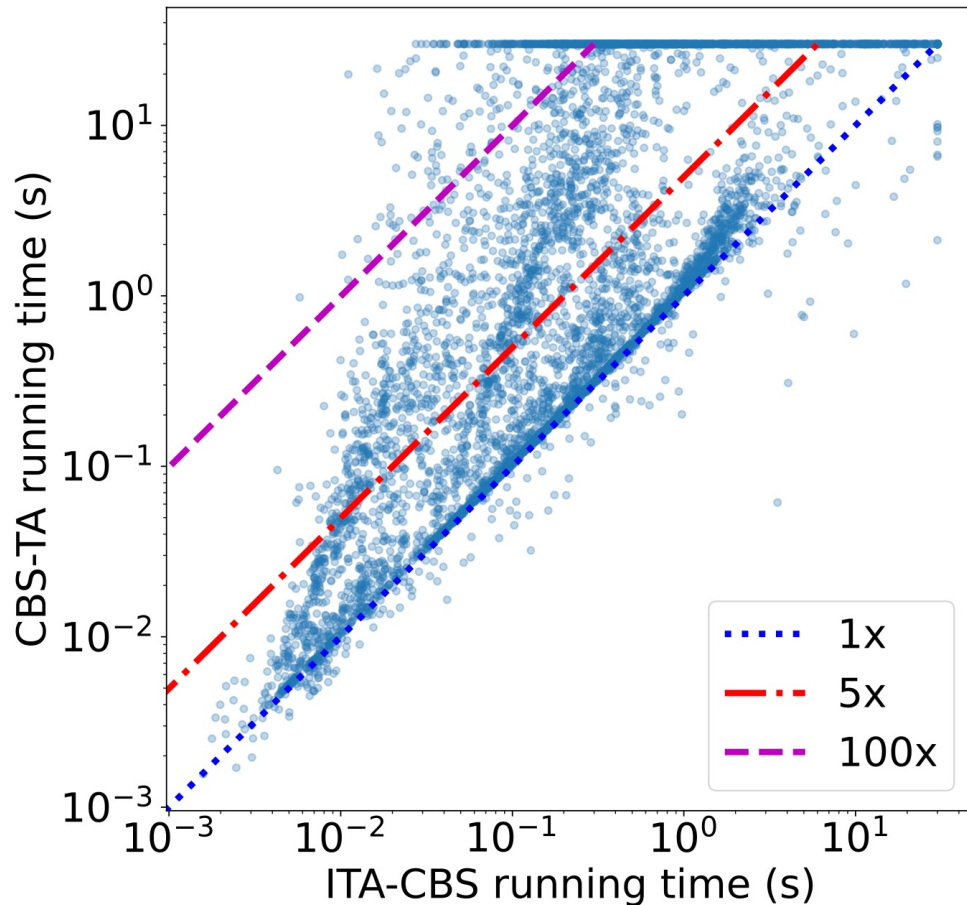
(d) Boston-0-256

Experiment Results

- Group test:
 - Each 5 agents forms a team
 - Each team shares 5 unique locations
- Common Target test:
 - Each agent's target set has **common locations shared by all agents**
 - Ratio of shared targets in each target set from 0%, 30%, 60%, to 100%
- #Agent: 15 — 60
- #target set size: 15 — 80 (depends on map size)

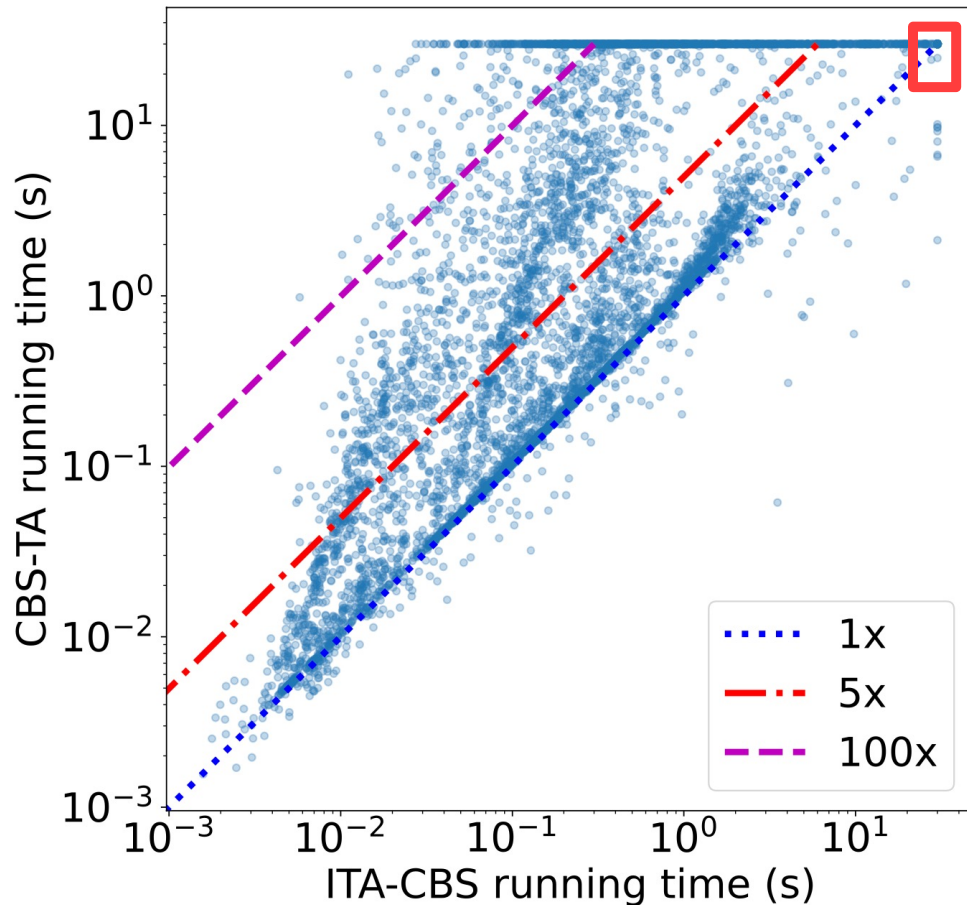
Experiment Results

- 7600 testcases



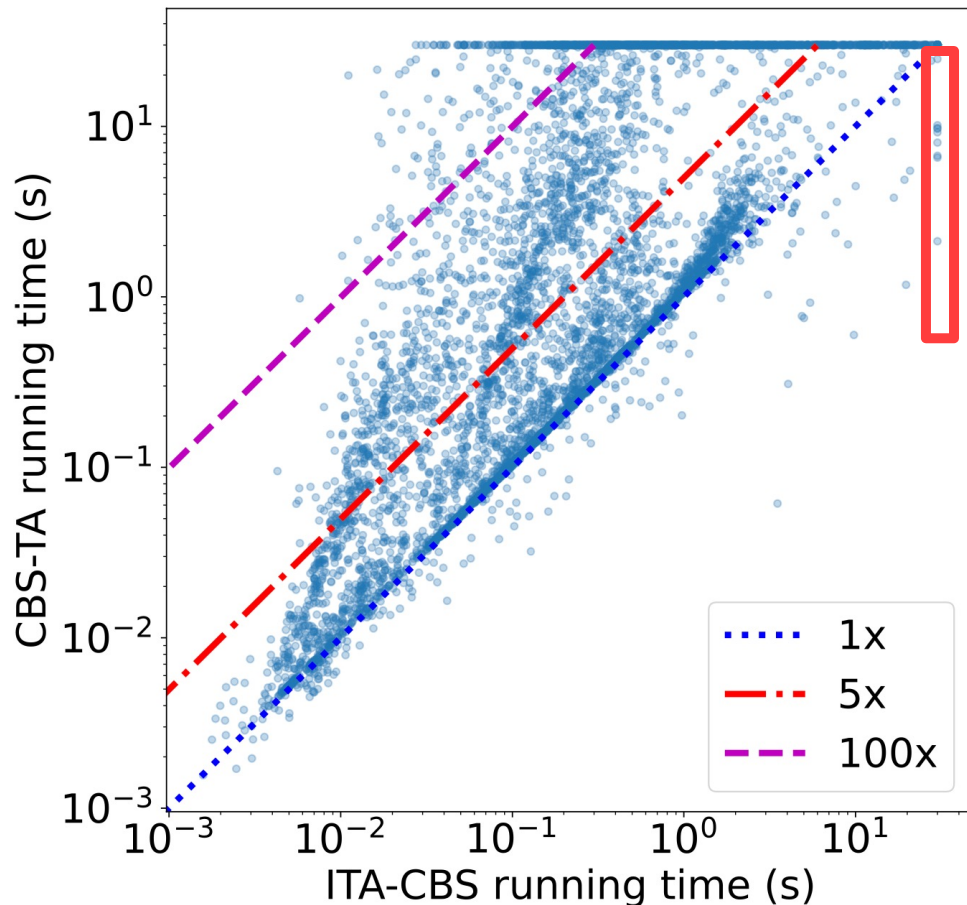
Experiment Results

- 7600 testcases
- 1266 both failed



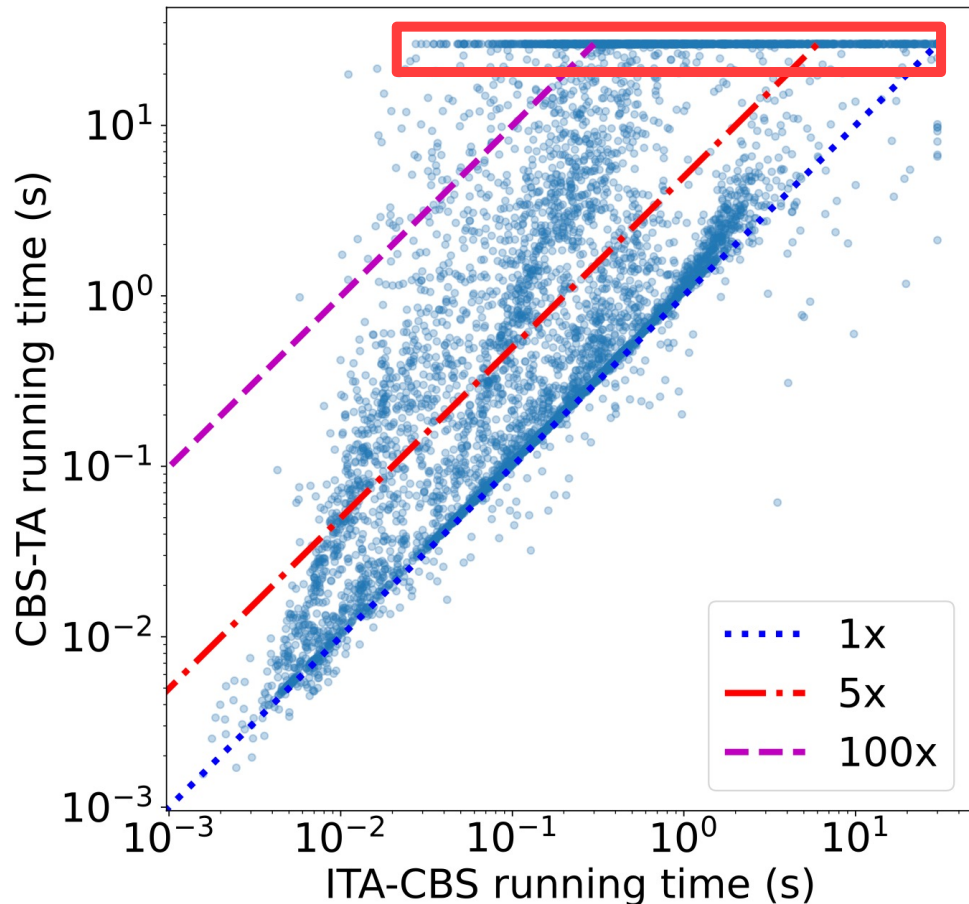
Experiment Results

- 7600 testcases
- 1266 both failed
- 9 CBS-TA solved only



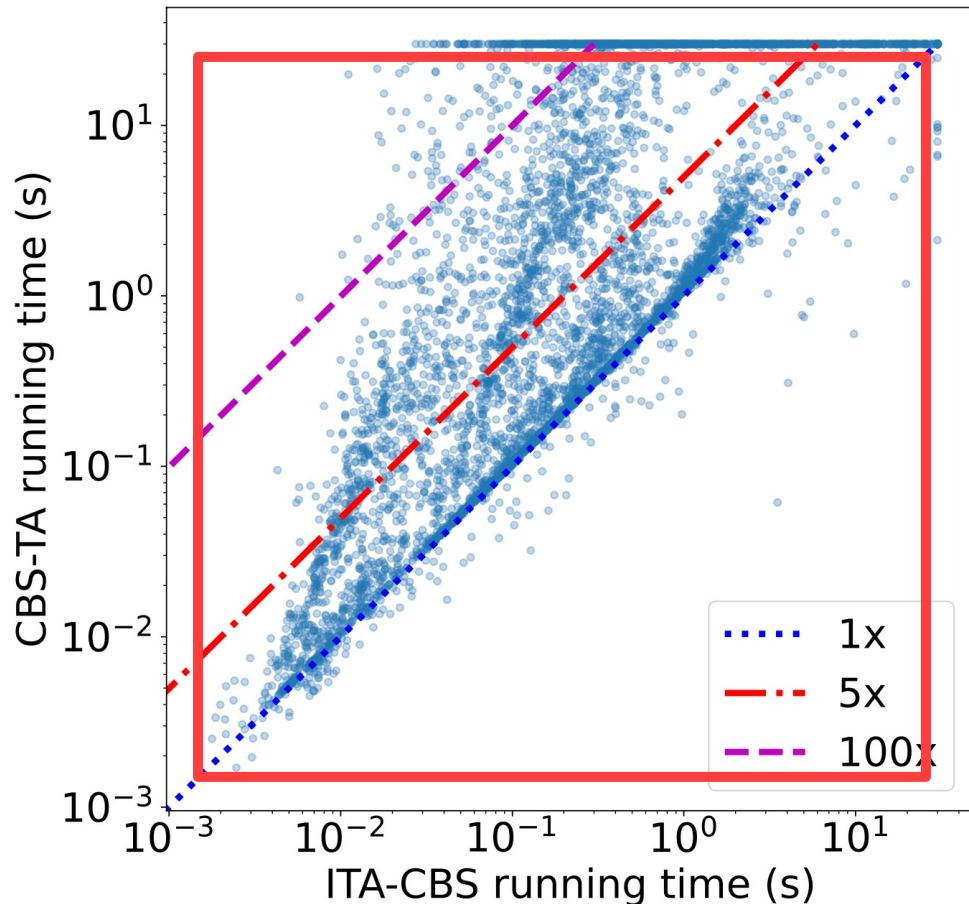
Experiment Results

- 7600 testcases
 - 1266 both failed
 - 9 CBS-TA solved only
 - 1191 ITA-CBS solved only



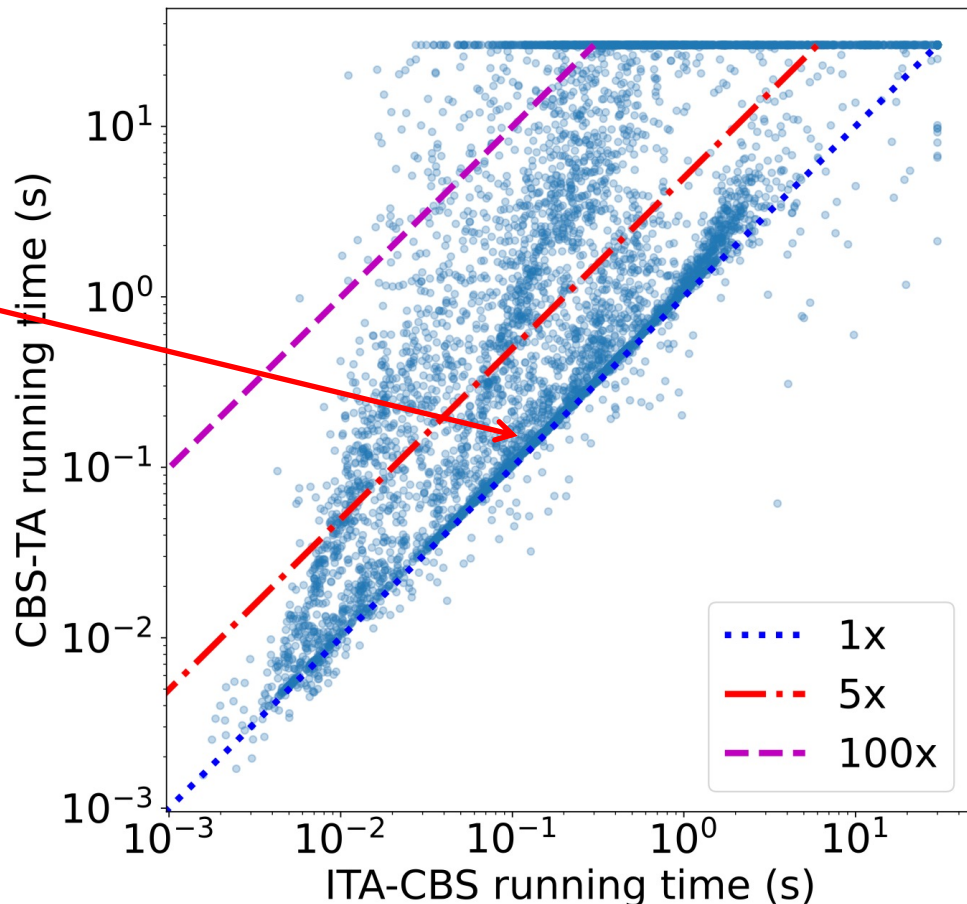
Experiment Results

- 7600 testcases
 - 1266 both failed
 - 9 CBS-TA solved only
 - 1191 ITA-CBS solved only
 - 5134 testcases both solved



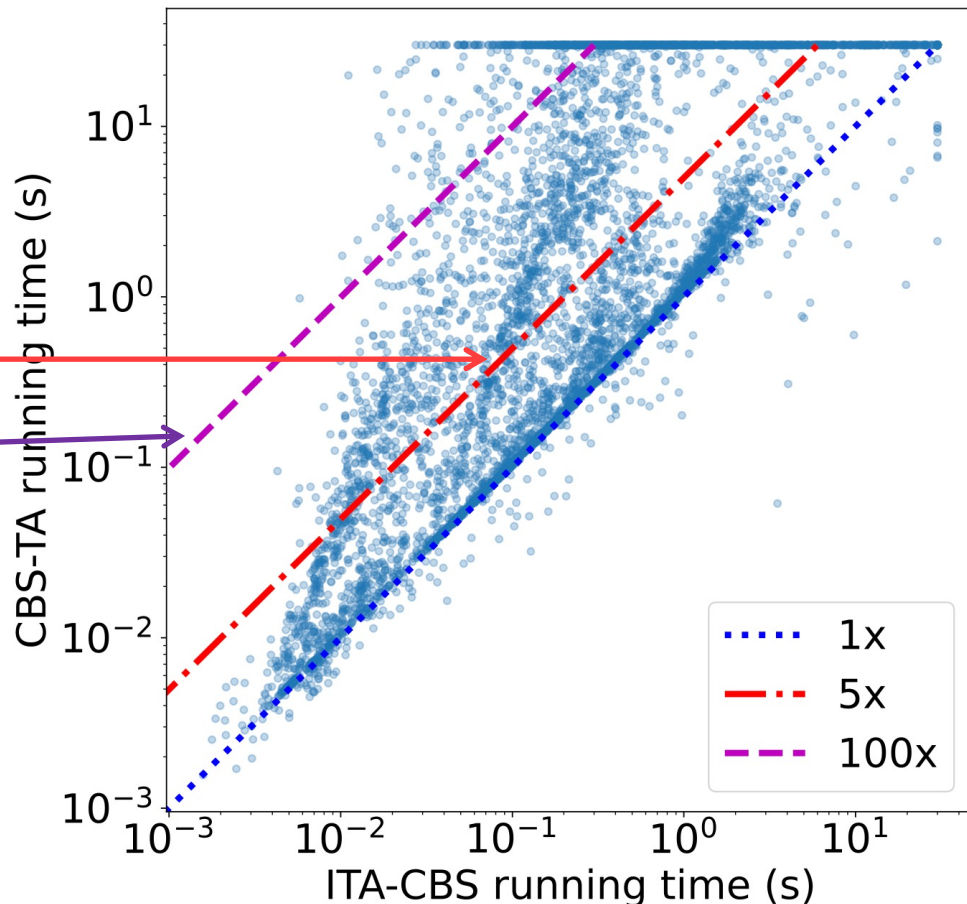
Experiment Results

- In 6334 (remove both failed)
 - 96.1% ITA-CBS 1x better
 - 38.7% 5x better
 - 5.6% 100x better



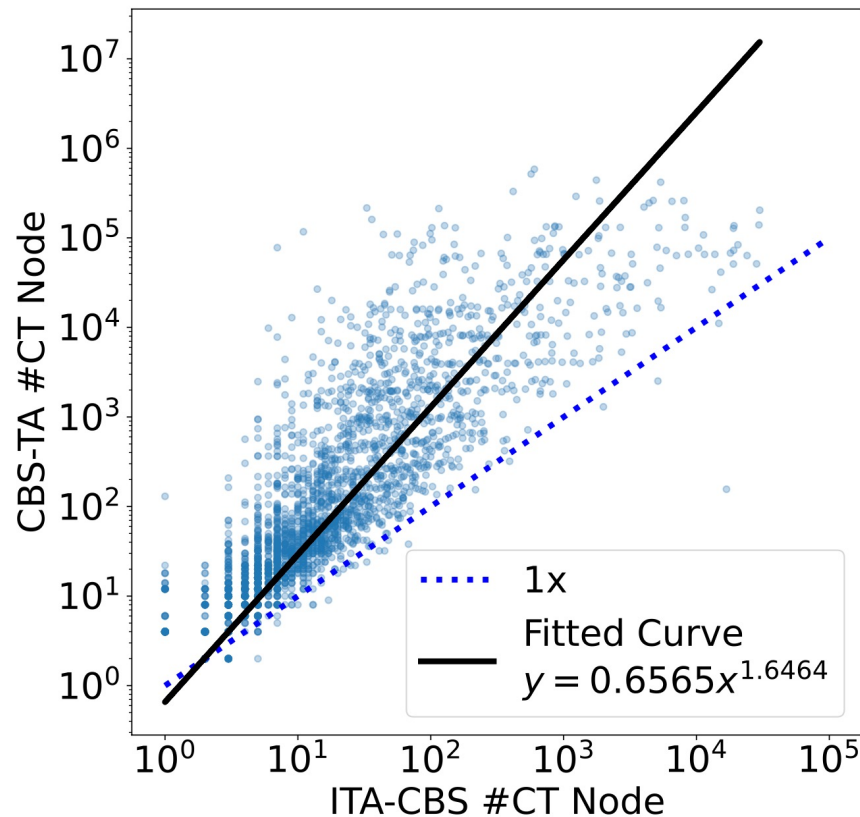
Experiment Results

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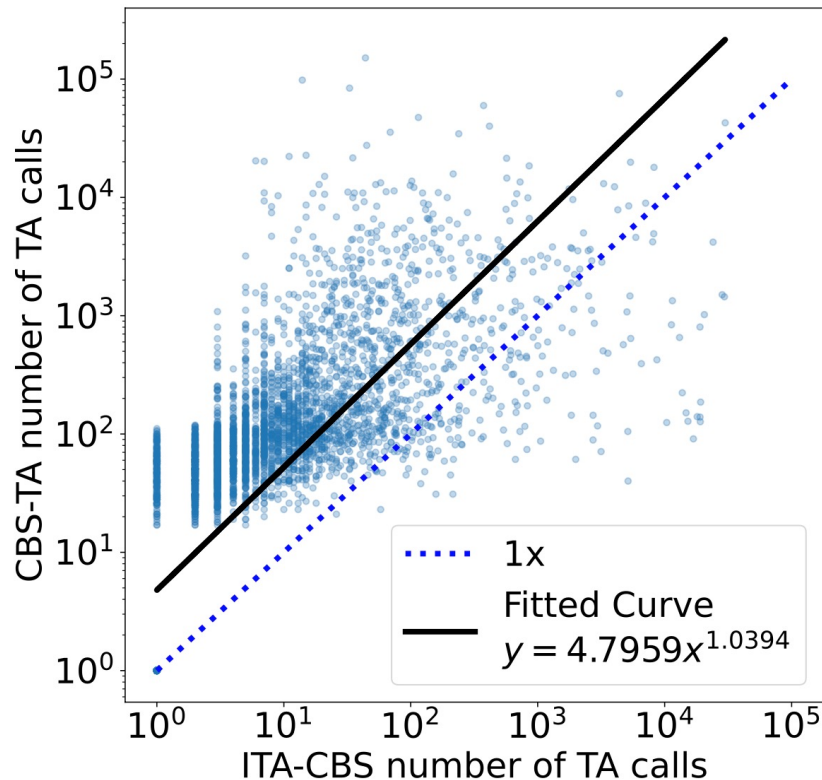
Experiment Results

- **Single Tree** structure makes ITA-CBS have **less nodes**



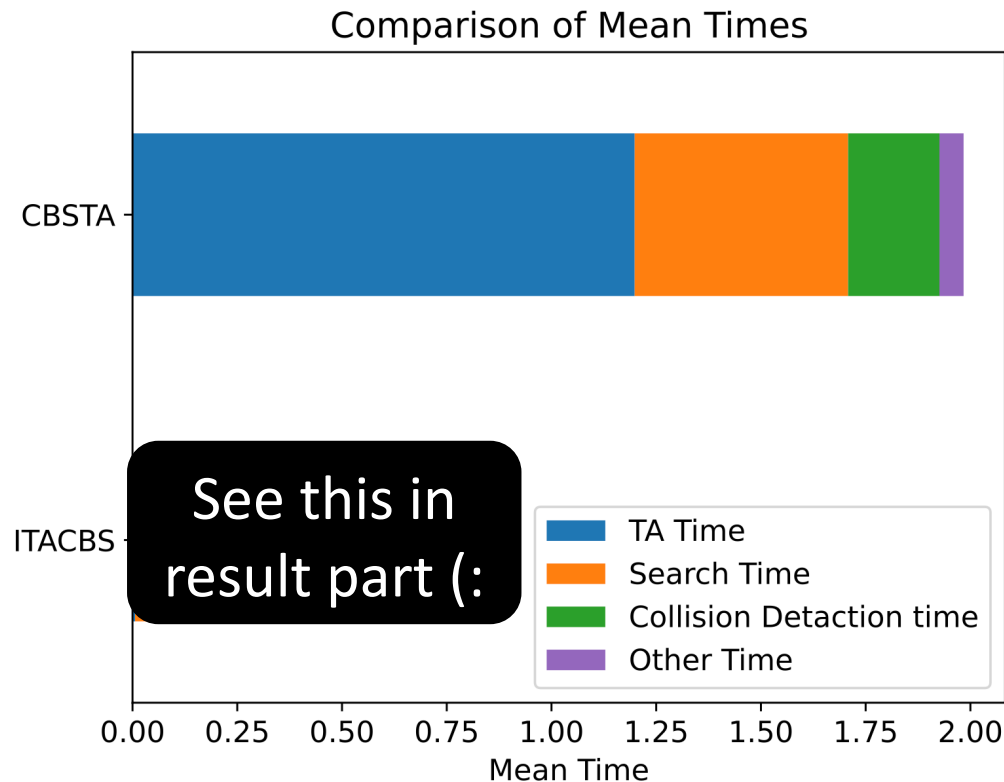
Experiment Results

- **Single Tree** structure makes ITA-CBS have **less nodes**
- **Single Tree** structure makes ITA-CBS have **less Task Assignment calls**



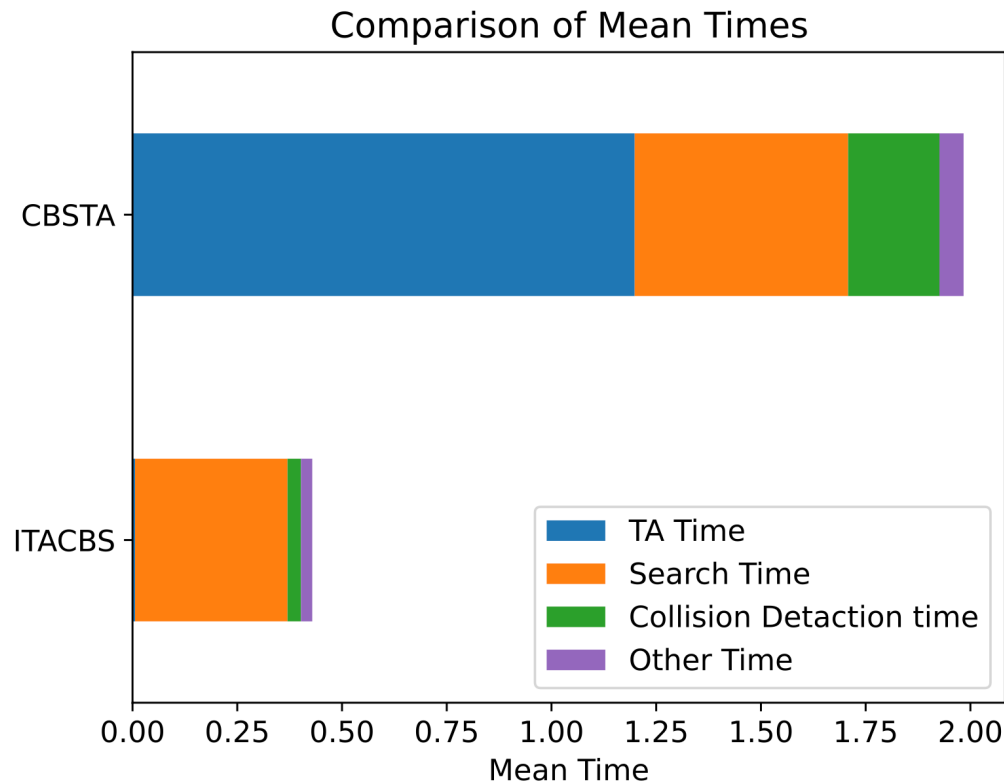
Experiment Results

- Half of the time for CBS-TA is spent on TA



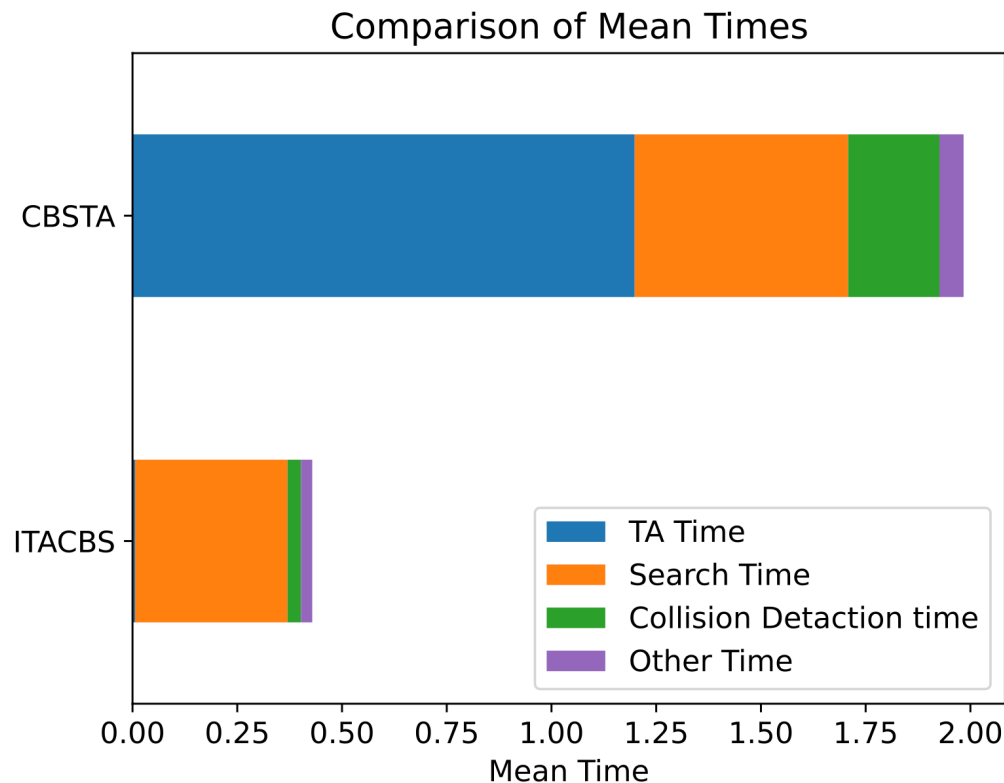
Experiment Results

- Half of the time for CBS-TA is spent on TA
- ITACBS has almost no proportion for TA time



Experiment Results

- Half of the time for CBS-TA is spent on TA
- ITACBS has almost no proportion for TA time
- Dynamic Hungarian algorithm saves lots of time for ITA-CBS



Compare CBS-TA and ITA-CBS

- CBS-TA (baseline):
 - Optimal
- ITA-CBS (ours):
 - Optimal

Compare CBS-TA and ITA-CBS

- CBS-TA (baseline):
 - Optimal
 - Constraint Tree Forest
- ITA-CBS (ours):
 - Optimal
 - Single Constraint Tree

Compare CBS-TA and ITA-CBS

- CBS-TA (baseline):
 - Optimal
 - Constraint Tree Forest
 - TA in Root CT node
- ITA-CBS (ours):
 - Optimal
 - Single Constraint Tree
 - TA in every CT node

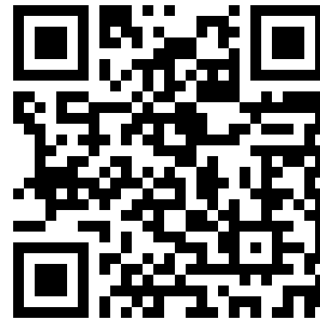
Compare CBS-TA and ITA-CBS

- CBS-TA (baseline):
 - Optimal
 - Constraint Tree Forest
 - TA in Root CT node
 - Next best TA is $O(N^3M)$
- ITA-CBS (ours):
 - Optimal
 - Single Constraint Tree
 - TA in every CT node
 - Incremental TA is $O(NM)$

Thank You!

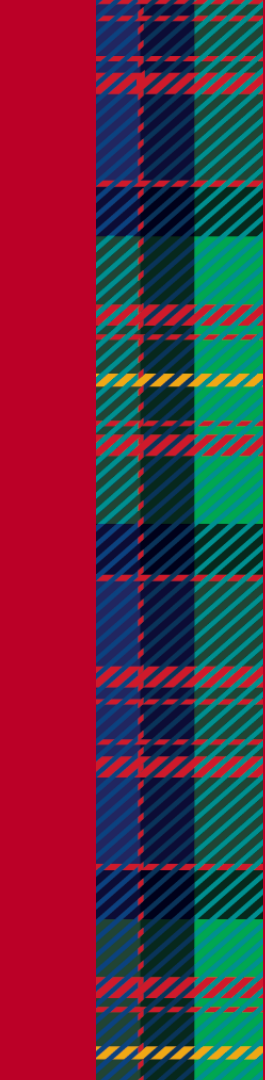
Paper:

https://sites.bu.edu/mrs2023/files/2023/11/0003_Fl.pdf



Code: <https://github.com/TachikakaMin/ITA-CBS2>





Carnegie Mellon University